MIAE 221: Materials Science

Master Formula Sheet Past Exam Solved Companion (Chapters 1–6)

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Strategic Overview & Pedagogical Standard

This master companion manual is the complete numerical and derivation reference for the **MIAE 221 Midterm Examination** covering **Chapters 1 through 6** (Callister 10th Edition: Atomic Structure & Interatomic Bonding, Crystal Structures & Crystallography, Imperfections & Point Defects, Diffusion, Mechanical Properties of Metals, and Dislocations/Strengthening).

For every formula in the curriculum, this guide provides an authentic Concordia past examination question, accompanied by:

1. **Zero-Skip Mathematical Derivation**: Every intermediate arithmetic and algebraic operation is explicitly displayed with SI dimensional units.

2. **"Why This Method Works"**: The underlying physical and crystallographic mechanism.

3. **Common Exam Pitfalls & Traps**: Diagnostic warnings detailing exactly where students lose marks on Concordia midterms.

Chapter 1: Atomic Structure & Interatomic Bonding

1.1 Total Atoms, Mass & Molar Conversion

$$\boxed{n_{\text{mol}} = \frac{m}{A} \qquad N_{\text{atoms}} = n_{\text{mol}} \cdot N_A = \frac{m}{A} N_A}$$

Past Exam Problem (Concordia Midterm)

A cylindrical wire of pure Platinum ($A_{\text{Pt}} = 195.08\ \text{g/mol}$, density $\rho_{\text{Pt}} = 21.45\ \text{g/cm}^3$) has a diameter of $d = 2.0\ \text{mm}$ and a length of $L = 1.50\ \text{m}$. Calculate (a) the total mass of the wire in grams, and (b) the total number of Platinum atoms contained within the specimen.

Zero-Skip Step-by-Step Solution

Step 1: Calculate Specimen Cross-Sectional Area and Volume

$$\text{Radius } r = \frac{d}{2} = \frac{2.0\ \text{mm}}{2} = 1.0\ \text{mm} = 0.10\ \text{cm}$$

$$\text{Length } L = 1.50\ \text{m} = 150.0\ \text{cm}$$

$$V = \pi r^2 L = \pi \times (0.10\ \text{cm})^2 \times 150.0\ \text{cm} = \pi \times 0.010\ \text{cm}^2 \times 150.0\ \text{cm} = 1.50\pi\ \text{cm}^3 = 4.71239\ \text{cm}^3$$

Step 2: Calculate Mass of the Specimen

$$m = \rho \times V = (21.45\ \text{g/cm}^3) \times (4.71239\ \text{cm}^3) = 101.0807\ \text{g} \approx \mathbf{101.08}\ \text{g}$$

Step 3: Calculate Moles and Total Atoms

$$n_{\text{mol}} = \frac{m}{A} = \frac{101.0807\ \text{g}}{195.08\ \text{g/mol}} = 0.51815\ \text{mol}$$

$$N_{\text{atoms}} = n_{\text{mol}} \times N_A = (0.51815\ \text{mol}) \times (6.022 \times 10^{23}\ \text{atoms/mol}) = \mathbf{3.120 \times 10^{23}}\ \text{atoms}$$

> **Why This Method Works**: Mass represents the macroscopic sum of atomic nuclei. Dividing by molar mass normalizes the sample into molar packets, which Avogadro's number scales into discrete atomic counts.

>

> **Exam Trap:** Mixing centimeters and millimeters. Always convert length and diameter to centimeters when working with density given in g/cm^3.

1.2 Percent Ionic Character (Pauling Electronegativity Relation)

$$\boxed{\%\text{Ionic Character} = \left[1 - \exp\left(-0.25\,(\chi_A - \chi_B)^2\right)\right] \times 100\%}$$

Past Exam Problem (Concordia Midterm)

Compute the percent ionic character of the interatomic bond in:

1. Silicon Carbide (SiC), given $\chi_{\text{Si}} = 1.8$ and $\chi_{\text{C}} = 2.5$.

2. Titanium Dioxide (TiO_2), given $\chi_{\text{Ti}} = 1.5$ and $\chi_{\text{O}} = 3.5$.

State whether each material is classified as predominantly ionic or covalent.

Zero-Skip Step-by-Step Solution

Part 1: Silicon Carbide (SiC)

1. Electronegativity difference: $\Delta\chi = |1.8 - 2.5| = 0.7$.

2. Square difference: $(\Delta\chi)^2 = (0.7)^2 = 0.49$.

3. Exponent: $-0.25 \times 0.49 = -0.1225$.

4. Evaluate exponential: $\exp(-0.1225) = 0.88471$.

5. Compute ionic fraction:

$$\%\text{IC} = [1 - 0.88471] \times 100\% = 0.11529 \times 100\% = \mathbf{11.53\%}$$

Classification: Because $\%\text{IC} < 50\%$, SiC is predominantly **covalent** (88.47% covalent).

Part 2: Titanium Dioxide (TiO_2)

1. Electronegativity difference: $\Delta\chi = |1.5 - 3.5| = 2.0$.

2. Square difference: $(\Delta\chi)^2 = (2.0)^2 = 4.0$.

3. Exponent: $-0.25 \times 4.0 = -1.00$.

4. Evaluate exponential: $\exp(-1.00) = 0.36788$.

5. Compute ionic fraction:

$$\%\text{IC} = [1 - 0.36788] \times 100\% = 0.63212 \times 100\% = \mathbf{63.21\%}$$

Classification: Because $\%\text{IC} > 50\%$, TiO_2 is predominantly **ionic** (63.21% ionic).

> **Why This Method Works**: Electronegativity quantifies an atom's affinity for bonding electrons. When $\Delta\chi > 1.7$, valence electrons are transferred rather than shared, generating localized electrostatic ionic bonds.

>

> **Exam Trap:** Never multiply $\Delta\chi$ by stoichiometric chemical subscripts (e.g., doubling for O_2). The Pauling equation calculates the character of an isolated pair bond between atom A and atom B.

1.3 Net Interatomic Force, Derivative & Equilibrium Spacing

$$\boxed{E_N(r) = -\frac{A}{r} + \frac{B}{r^n} \qquad F_N(r) = \frac{dE_N}{dr} = \frac{A}{r^2} - \frac{nB}{r^{n+1}} = 0 \quad \text{at } r = r_0}$$

$$\boxed{r_0 = \left(\frac{nB}{A}\right)^{\frac{1}{n-1}} \qquad E_0 = -\frac{A}{r_0}\left(1 - \frac{1}{n}\right)}$$

Past Exam Problem (Concordia Midterm)

The attractive and repulsive bonding energies between two isolated ions are given by:

$$E_A(r) = -\frac{1.436}{r}\ \text{eV} \qquad E_R(r) = \frac{7.32 \times 10^{-6}}{r^8}\ \text{eV} \quad (r \text{ in nm})$$

1. Determine the equilibrium interionic separation r_0 in nanometers (nm).

2. Calculate the bonding energy E_0 in electron-volts (eV) and Joules (J).

Zero-Skip Step-by-Step Solution

Step 1: Set Up Net Potential Energy & Derivative

$$E_N(r) = -1.436 r^{-1} + 7.32 \times 10^{-6} r^{-8}$$

$$\frac{dE_N}{dr} = (-1)(-1.436)r^{-2} + (-8)(7.32 \times 10^{-6})r^{-9} = \frac{1.436}{r^2} - \frac{5.856 \times 10^{-5}}{r^9}$$

Step 2: Solve for Equilibrium Separation r_0

$$\frac{1.436}{r_0^2} = \frac{5.856 \times 10^{-5}}{r_0^9} \implies r_0^7 = \frac{5.856 \times 10^{-5}}{1.436} = 4.07799 \times 10^{-5}\ \text{nm}^7$$

$$r_0 = (4.07799 \times 10^{-5})^{1/7} = \mathbf{0.23597}\ \text{nm} \approx \mathbf{0.236}\ \text{nm}$$

Step 3: Evaluate Bonding Energy E_0

$$E_A(r_0) = -\frac{1.436}{0.23597} = -6.0855 \text{ eV}$$

$$E_R(r_0) = \frac{7.32 \times 10^{-6}}{(0.23597)^8} = \frac{7.32 \times 10^{-6}}{9.6229 \times 10^{-6}} = +0.7607 \text{ eV}$$

$$E_0 = E_A(r_0) + E_R(r_0) = -6.0855 + 0.7607 = \mathbf{-5.325} \text{ eV}$$

Convert to Joules ($1 \text{ eV} = 1.602 \times 10^{-19}$ J):

$$E_0 = (-5.3248 \text{ eV}) \times (1.602 \times 10^{-19} \text{ J/eV}) = \mathbf{-8.530 \times 10^{-19}} \text{ J}$$

> **Why This Method Works**: At the bottom of the potential energy trough, the slope $dE/dr = 0$, meaning attractive and repulsive forces cancel exactly. The minimum energy well depth E_0 defines the cohesive bond energy.

>

> **Exam Trap:** Setting $E_N(r) = 0$ instead of $dE/dr = 0$. $E_N = 0$ is the crossover point where net energy vanishes, not the equilibrium bond distance.

Chapter 2: Crystal Structures & Crystallography

2.1 Theoretical Density of Metals

$$\boxed{\rho = \frac{nA}{V_C N_A} = \frac{nA}{a^3 N_A}}$$

Past Exam Problem (Concordia Midterm)

Copper (Cu) has an FCC crystal structure, an atomic radius of $R = 0.1278$ nm, and an atomic weight of $A = 63.55$ g/mol. Calculate the theoretical density of Cu in g/cm^3 and compare it with the handbook experimental density of 8.94 g/cm^3.

Zero-Skip Step-by-Step Solution

Step 1: Convert Radius to Centimeters & Find Lattice Parameter a

$$R = 0.1278 \text{ nm} = 0.1278 \times 10^{-7} \text{ cm} = 1.278 \times 10^{-8} \text{ cm}$$

For FCC, atoms touch along the face diagonal: $4R = a\sqrt{2}$:

$$a = 2R\sqrt{2} = 2 \times (1.278 \times 10^{-8} \text{ cm}) \times 1.414214 = 3.61473 \times 10^{-8} \text{ cm}$$

Step 2: Calculate Unit Cell Volume V_C

$$V_C = a^3 = (3.61473 \times 10^{-8} \text{ cm})^3 = 4.72314 \times 10^{-23} \text{ cm}^3$$

Step 3: Calculate Density

For FCC Cu, $n = 4$ atoms/cell:

$$\text{Numerator: } n \times A = 4 \times 63.55 \text{ g/mol} = 254.20 \text{ g/mol}$$

$$\text{Denominator: } V_C \times N_A = (4.72314 \times 10^{-23} \text{ cm}^3) \times (6.022 \times 10^{23} \text{ mol}^{-1}) = 28.4427 \text{ cm}^3/\text{mol}$$

$$\rho = \frac{254.20 \text{ g/mol}}{28.4427 \text{ cm}^3/\text{mol}} = \mathbf{8.937} \text{ g/cm}^{\mathbf{3}}$$

$$\% \text{ difference} = \frac{|8.937 - 8.94|}{8.94} \times 100\% = \mathbf{0.034\%}$$

> **Why This Method Works**: A macroscopic single crystal is an infinite tessellation of identical unit cells. Thus, bulk macroscopic density equals the density of a single unit cell.

>

> **Exam Trap:** Forgetting that $1 \text{ nm}^3 = 10^{-21} \text{ cm}^3$. If you compute a in meters, density outputs in kg/m^3 (8937 kg/m^3). Divide by 1000 to get g/cm^3.

2.2 Total Unit Cells in a Solid Sample

$$\boxed{N_{\text{cells}} = \frac{V_{\text{sample}}}{V_C} = \frac{m/\rho}{a^3}}$$

Past Exam Problem (Concordia Midterm)

Calculate the total number of unit cells in a solid 1.0 cm^3 cube of pure BCC Iron (α-Fe) having a lattice parameter of $a = 0.2866 \text{ nm}$.

Zero-Skip Step-by-Step Solution

Step 1: Calculate the Volume of a Single BCC Unit Cell

$$a = 0.2866 \text{ nm} = 0.2866 \times 10^{-7} \text{ cm} = 2.866 \times 10^{-8} \text{ cm}$$

$$V_C = a^3 = (2.866 \times 10^{-8} \text{ cm})^3 = 2.3541 \times 10^{-23} \text{ cm}^3$$

Step 2: Divide Sample Volume by Cell Volume

$$V_{\text{sample}} = 1.0 \text{ cm}^3$$

$$N_{\text{cells}} = \frac{1.0 \text{ cm}^3}{2.3541 \times 10^{-23} \text{ cm}^3} = \mathbf{4.248 \times 10^{22}} \text{ unit cells}$$

> **Exam Trap:** Confusing total unit cells (N_{cells}) with total atoms (N_{atoms}). For BCC, each cell has 2 atoms, so $N_{\text{atoms}} = 2 \times N_{\text{cells}} = 8.496 \times 10^{22}$ atoms.

2.3 APF & Geometric Derivations ($a \leftrightarrow R$)

$$\boxed{\text{APF} = \frac{V_{\text{atoms}}}{V_C} = \frac{n\left(\frac{4}{3}\pi R^3\right)}{V_C}}$$

Past Exam Problem (Concordia Midterm)

Derive the expression relating the unit cell edge length a to the atomic radius R for BCC Tungsten ($R = 0.1371\ \text{nm}$). Prove from first principles that the APF for BCC is identically 0.68.

Zero-Skip Step-by-Step Solution

Step 1: Geometric Derivation of $a(R)$

- Body diagonal passes through corners and center atom: $\text{Length} = R + 2R + R = 4R$.
- Face diagonal $f = \sqrt{a^2 + a^2} = a\sqrt{2}$.
- Body diagonal $b = \sqrt{f^2 + a^2} = \sqrt{2a^2 + a^2} = a\sqrt{3}$.

$$a\sqrt{3} = 4R \implies \mathbf{a = \frac{4R}{\sqrt{3}}}$$

Step 2: Compute a and V_C for Tungsten

$$a = \frac{4 \times 0.1371\ \text{nm}}{\sqrt{3}} = \frac{0.5484}{1.73205} = 0.31662\ \text{nm} = 3.1662 \times 10^{-10}\ \text{m}$$

$$V_C = a^3 = (3.1662 \times 10^{-10}\ \text{m})^3 = \mathbf{3.174 \times 10^{-29}\ m^3}$$

Step 3: Algebraic Proof of $\text{APF} = 0.68$

$$n_{\text{BCC}} = 2\ \text{atoms}, \quad V_S = 2 \times \left(\frac{4}{3}\pi R^3\right) = \frac{8}{3}\pi R^3$$

$$V_C = \left(\frac{4R}{\sqrt{3}}\right)^3 = \frac{64R^3}{3\sqrt{3}}$$

$$\text{APF} = \frac{\frac{8}{3}\pi R^3}{\frac{64R^3}{3\sqrt{3}}} = \frac{8\pi}{3} \times \frac{3\sqrt{3}}{64} = \frac{\sqrt{3}\pi}{8} = \frac{1.73205 \times 3.14159}{8} = \frac{5.4414}{8} = \mathbf{0.6802 \approx 0.68}$$

2.4 Linear Density (LD) & Planar Density (PD)

$$\boxed{\text{LD} = \frac{\text{Atoms centered on direction vector}}{\text{Length of vector } L} \qquad \text{PD} = \frac{\text{Atoms centered on plane}}{\text{Area of plane } A_p}}$$

Past Exam Problem (Concordia Midterm)

1. Calculate the linear density of the $[110]$ direction in FCC Aluminum ($R = 0.143\ \text{nm}$).
2. Calculate the planar density of the (110) plane in BCC Iron ($a = 0.2866\ \text{nm}$).

Zero-Skip Step-by-Step Solution

Part 1: Linear Density of $[110]$ in FCC

- In FCC, $[110]$ runs along the face diagonal: $L = 4R = 4 \times 0.143 \text{ nm} = 0.572 \text{ nm}$.

- Intersected atoms: $\frac{1}{2}$ (corner) $+ 1$ (face-center) $+ \frac{1}{2}$ (corner) $= 2$ atoms.

$$\text{LD}_{[110]} = \frac{2 \text{ atoms}}{0.572 \text{ nm}} = \mathbf{3.50} \text{ nm}^{-1} = \mathbf{3.50 \times 10^{9}} \text{ m}^{-1}$$

Part 2: Planar Density of (110) in BCC

- Dimensions of (110) rectangular plane: Width $= a$, Height $= a\sqrt{2}$.

$$A_p = a \times a\sqrt{2} = a^2\sqrt{2} = (0.2866 \text{ nm})^2 \times 1.414214 = 0.08214 \times 1.414214 = 0.11616 \text{ nm}^2$$

- Intersected atoms: 4 corners $\times \frac{1}{4} + 1$ center $= 2$ atoms.

$$\text{PD}_{(110)} = \frac{2 \text{ atoms}}{0.11616 \text{ nm}^2} = \mathbf{17.22} \text{ nm}^{-2} = \mathbf{1.72 \times 10^{19}} \text{ m}^{-2}$$

2.5 Interplanar Spacing (d_{hkl}) & Bragg's Law

$$\boxed{d_{hkl} = \frac{a}{\sqrt{h^2 + k^2 + l^2}} \qquad n\lambda = 2d_{hkl}\sin\theta}$$

Diffraction Rules: BCC: $h + k + l =$ even FCC: h, k, l all odd or all even

Past Exam Problem (Concordia Midterm)

A monochromatic X-ray beam ($\lambda = 0.1542$ nm) reflects from the (110) planes of a BCC metal at a two-theta diffraction angle of $2\theta = 44.50°$. Calculate (a) d_{110}, (b) the unit cell parameter a, and (c) the atomic radius R.

Zero-Skip Step-by-Step Solution

Step 1: Bragg Angle & Interplanar Spacing

$$\theta = \frac{44.50°}{2} = 22.25° \implies \sin(22.25°) = 0.37865$$

$$d_{110} = \frac{\lambda}{2\sin\theta} = \frac{0.1542 \text{ nm}}{2 \times 0.37865} = \frac{0.1542}{0.75730} = \mathbf{0.20362} \text{ nm}$$

Step 2: Lattice Parameter a

$$d_{110} = \frac{a}{\sqrt{1^2 + 1^2 + 0^2}} = \frac{a}{\sqrt{2}} \implies a = d_{110}\sqrt{2} = 0.20362 \text{ nm} \times 1.414214 = \mathbf{0.28796} \text{ nm} \approx \mathbf{0.288} \text{ nm}$$

Step 3: Atomic Radius R for BCC

$$a = \frac{4R}{\sqrt{3}} \implies R = \frac{a\sqrt{3}}{4} = \frac{0.28796 \text{ nm} \times 1.73205}{4} = \mathbf{0.12469} \text{ nm} \approx \mathbf{0.125} \text{ nm}$$

> 🚨 **Exam Trap:** Diffractometers output **2θ**. Always divide by 2 to obtain the Bragg angle θ before computing $\sin\theta$.

Chapter 3: Imperfections in Solids

3.1 Equilibrium Vacancy Concentration & Atomic Site Density

$$\boxed{N_v = N \exp\left(-\frac{Q_v}{kT}\right) \qquad N = \frac{\rho N_A}{A}}$$

Past Exam Problem (Concordia Midterm)

Calculate the equilibrium concentration of vacancies per cubic meter in Copper at 850°C ($1123.15\ \text{K}$).

Given: $Q_v = 0.90\ \text{eV/atom}$, $\rho = 8.40\ \text{g/cm}^3 = 8.40 \times 10^6\ \text{g/m}^3$, $A = 63.55\ \text{g/mol}$, $k = 8.62 \times 10^{-5}\ \text{eV/atom} \cdot \text{K}$, $N_A = 6.022 \times 10^{23}\ \text{atoms/mol}$.

Zero-Skip Step-by-Step Solution

Step 1: Calculate Total Atomic Sites per m^3 (N)

$$N = \frac{(8.40 \times 10^6\ \text{g/m}^3) \times (6.022 \times 10^{23}\ \text{atoms/mol})}{63.55\ \text{g/mol}} = \frac{5.05848 \times 10^{30}}{63.55} = 7.95984 \times 10^{28}\ \text{atoms/m}^3$$

Step 2: Evaluate the Exponential Factor

$$kT = (8.62 \times 10^{-5}\ \text{eV/K}) \times (1123.15\ \text{K}) = 0.0968155\ \text{eV}$$

$$-\frac{Q_v}{kT} = -\frac{0.90\ \text{eV}}{0.0968155\ \text{eV}} = -9.29603$$

$$\exp(-9.29603) = 9.1788 \times 10^{-5}$$

Step 3: Calculate N_v

$$N_v = (7.95984 \times 10^{28}\ \text{atoms/m}^3) \times (9.1788 \times 10^{-5}) = \mathbf{7.306 \times 10^{24}}\ \text{vacancies/m}^3$$

> **Exam Trap:** Pairing Q_v in eV with the gas constant $R = 8.314\ \text{J/mol} \cdot \text{K}$. If Q_v is in eV/atom, use $k = 8.62 \times 10^{-5}\ \text{eV/atom} \cdot \text{K}$. If Q_v is in J/mol, use $R = 8.314\ \text{J/mol} \cdot \text{K}$.

3.2 Composition Conversions & Average Alloy Density

$$\boxed{C'_1 = \frac{C_1/A_1}{\frac{C_1}{A_1} + \frac{C_2}{A_2}} \times 100\% \qquad \rho_{\text{avg}} = \frac{100}{\frac{C_1}{\rho_1} + \frac{C_2}{\rho_2}}}$$

Past Exam Problem (Concordia Midterm)

An alloy contains $97.0\ \text{wt\%}$ Al ($A = 26.98\ \text{g/mol}$) and $3.0\ \text{wt\%}$ Cu ($A = 63.55\ \text{g/mol}$). Convert this composition into atomic percent ($\text{at\%}$). Furthermore, calculate the theoretical density of brass ($70.0\ \text{wt\%}$ Cu, $\rho = 8.94\ \text{g/cm}^3$; $30.0\ \text{wt\%}$ Zn, $\rho = 7.13\ \text{g/cm}^3$).

Zero-Skip Step-by-Step Solution

Part 1: $\mathrm{wt}\%$ to $\mathrm{at}\%$ Conversion

$$\text{Moles Al} = \frac{97.0}{26.98} = 3.59526 \text{ mol} \qquad \text{Moles Cu} = \frac{3.0}{63.55} = 0.04721 \text{ mol}$$

$$\text{Total moles} = 3.59526 + 0.04721 = 3.64247 \text{ mol}$$

$$C'_{\text{Cu}} = \frac{0.04721}{3.64247} \times 100\% = \mathbf{1.30} \text{ at}\% \qquad C'_{\text{Al}} = 100\% - 1.30\% = \mathbf{98.70} \text{ at}\%$$

Part 2: Average Density Calculation

$$\frac{C_{\text{Cu}}}{\rho_{\text{Cu}}} = \frac{70.0}{8.94} = 7.82998 \text{ cm}^3 \qquad \frac{C_{\text{Zn}}}{\rho_{\text{Zn}}} = \frac{30.0}{7.13} = 4.20757 \text{ cm}^3$$

$$\rho_{\text{avg}} = \frac{100}{7.82998 + 4.20757} = \frac{100}{12.03755} = \mathbf{8.307} \text{ g/cm}^{\mathbf{3}}$$

> 🚨 **Exam Trap:** Never take the weighted arithmetic mean $\rho = 0.70(8.94) + 0.30(7.13) = 8.397 \text{ g/cm}^3$ (WRONG). Mass fractions cannot weight volume directly; the reciprocal harmonic mean is required.

3.3 ASTM Grain Size Number

$$\boxed{N = 2^{n-1} \qquad N_M \left(\frac{M}{100}\right)^2 = 2^{n-1} \implies n = 1 + \frac{\ln\left[N_M (M/100)^2\right]}{\ln 2}}$$

Past Exam Problem (Concordia Midterm)

A micrograph taken at $250\times$ magnification reveals 40 grains in an area of exactly 1 square inch. Determine the ASTM grain size number n.

Zero-Skip Step-by-Step Solution

Step 1: Scale to Standard $100\times$ Magnification

$$N = N_M \left(\frac{M}{100}\right)^2 = 40 \times \left(\frac{250}{100}\right)^2 = 40 \times (2.5)^2 = 40 \times 6.25 = 250 \text{ grains/in}^2$$

Step 2: Solve for n Using Logarithms

$$250 = 2^{n-1} \implies \ln(250) = (n-1)\ln(2)$$

$$n - 1 = \frac{\ln(250)}{\ln(2)} = \frac{5.52146}{0.69315} = 7.9658 \implies n = 7.9658 + 1 = \mathbf{8.97} \approx \mathbf{9.0}$$

Chapter 4: Diffusion

4.1 Fick's First Law (Steady-State Flux) & Arrhenius Temperature Scaling

$$\boxed{J = -D\frac{dC}{dx} = D\frac{\Delta C}{\Delta x} \qquad D = D_0 \exp\left(-\frac{Q_d}{RT}\right)}$$

$$\boxed{\ln\left(\frac{D_2}{D_1}\right) = \frac{Q_d}{R}\left(\frac{1}{T_1} - \frac{1}{T_2}\right)}$$

Past Exam Problem (Concordia Midterm)

1. A sheet of BCC iron $1.5\ \text{mm}$ thick has surface carbon concentrations of $C_A = 1.4\ \text{kg/m}^3$ and $C_B = 0.2\ \text{kg/m}^3$. At 700°C, $D = 1.2 \times 10^{-10}\ \text{m}^2/\text{s}$. Calculate the steady-state diffusion flux J.

2. The diffusion coefficient of Zn in Cu is $5.0 \times 10^{-15}\ \text{m}^2/\text{s}$ at 500°C ($773.15\ \text{K}$) and $2.0 \times 10^{-13}\ \text{m}^2/\text{s}$ at 650°C ($923.15\ \text{K}$). Calculate Q_d in kJ/mol and D_0 in m^2/s.

Zero-Skip Step-by-Step Solution

Part 1: Steady-State Flux

$$\frac{\Delta C}{\Delta x} = \frac{1.4 - 0.2\ \text{kg/m}^3}{1.5 \times 10^{-3}\ \text{m}} = \frac{1.2}{1.5 \times 10^{-3}} = 800\ \text{kg/m}^4$$

$$J = D\left(\frac{\Delta C}{\Delta x}\right) = (1.2 \times 10^{-10}\ \text{m}^2/\text{s}) \times (800\ \text{kg/m}^4) = \mathbf{9.60 \times 10^{-8}}\ \text{kg/(m}^\mathbf{2}\cdot\text{s)}$$

Part 2: Arrhenius Parameters

$$\ln\left(\frac{D_2}{D_1}\right) = \ln\left(\frac{2.0 \times 10^{-13}}{5.0 \times 10^{-15}}\right) = \ln(40) = 3.68888$$

$$\frac{1}{T_1} - \frac{1}{T_2} = \frac{1}{773.15} - \frac{1}{923.15} = 0.0012934 - 0.0010833 = 2.1016 \times 10^{-4}\ \text{K}^{-1}$$

$$Q_d = \frac{3.68888 \times 8.314\ \text{J/mol}\cdot\text{K}}{2.1016 \times 10^{-4}\ \text{K}^{-1}} = \frac{30.6723}{2.1016 \times 10^{-4}} = 145947\ \text{J/mol} = \mathbf{145.95}\ \text{kJ/mol}$$

Now solve for D_0:

$$D_0 = \frac{D_1}{\exp(-Q_d/RT_1)} = \frac{5.0 \times 10^{-15}}{\exp\left(-\frac{145947}{8.314 \times 773.15}\right)} = \frac{5.0 \times 10^{-15}}{\exp(-22.705)} = \frac{5.0 \times 10^{-15}}{1.3782 \times 10^{-10}} = \mathbf{3.63 \times 10^{-5}}\ \text{m}^\mathbf{2}/\text{s}$$

4.2 Fick's Second Law (Carburization & Error Function Interpolation)

$$\boxed{\frac{C_x - C_0}{C_s - C_0} = 1 - \text{erf}\left(\frac{x}{2\sqrt{Dt}}\right) \qquad \frac{x_1^2}{D_1 t_1} = \frac{x_2^2}{D_2 t_2} = \text{const}}$$

Past Exam Problem (Concordia Final Exam)

Steel with initial carbon $C_0 = 0.20\ \text{wt}\%$ is carburized at 950°C ($D = 1.6 \times 10^{-11}\ \text{m}^2/\text{s}$) with surface carbon $C_s = 1.00\ \text{wt}\%$.

1. Calculate the time t (in hours) to achieve $C_x = 0.60\ \text{wt}\%$ at a case depth of $x = 0.75\ \text{mm}$.

Given: $\text{erf}(0.45) = 0.4755$, $\text{erf}(0.50) = 0.5205$.

2. At the same temperature, how many hours are required to reach the same carbon concentration at depth $x = 1.50\ \text{mm}$?

Zero-Skip Step-by-Step Solution

Part 1: Non-Steady-State Carburization Time

$$\frac{C_x - C_0}{C_s - C_0} = \frac{0.60 - 0.20}{1.00 - 0.20} = \frac{0.40}{0.80} = 0.5000$$

$$1 - \text{erf}(z) = 0.5000 \implies \text{erf}(z) = 0.5000$$

Interpolate for z:

$$z = 0.45 + \frac{0.5000 - 0.4755}{0.5205 - 0.4755} \times (0.50 - 0.45) = 0.45 + \frac{0.0245}{0.0450} \times 0.05 = 0.45 + 0.02722 = 0.47722$$

Now invert $z = \frac{x}{2\sqrt{Dt}}$:

$$\sqrt{Dt} = \frac{x}{2z} = \frac{0.75 \times 10^{-3}\ \text{m}}{2 \times 0.47722} = \frac{7.5 \times 10^{-4}}{0.95444} = 7.8580 \times 10^{-4}\ \text{m}$$

$$Dt = (7.8580 \times 10^{-4})^2 = 6.1748 \times 10^{-7}\ \text{m}^2$$

$$t = \frac{6.1748 \times 10^{-7}\ \text{m}^2}{1.6 \times 10^{-11}\ \text{m}^2/\text{s}} = 38592.5\ \text{seconds}$$

$$t_{\text{hours}} = \frac{38592.5}{3600} = \mathbf{10.72}\ \text{hours}$$

Part 2: Depth Scaling at Constant Temperature

$$\frac{x_1^2}{t_1} = \frac{x_2^2}{t_2} \implies t_2 = t_1 \left(\frac{x_2}{x_1}\right)^2 = (10.72\ \text{h}) \times \left(\frac{1.50\ \text{mm}}{0.75\ \text{mm}}\right)^2 = 10.72 \times (2)^2 = 10.72 \times 4 = \mathbf{42.88}\ \text{hours}$$

Chapter 5: Mechanical Properties of Metals

5.1 Engineering Stress, Strain, Modulus of Elasticity & Poisson's Ratio

$$\boxed{\sigma = \frac{F}{A_0} \qquad \epsilon_z = \frac{\Delta l}{l_0} \qquad E = \frac{\sigma}{\epsilon_z} \qquad \nu = -\frac{\epsilon_x}{\epsilon_z} \qquad G = \frac{E}{2(1+\nu)}}$$

Past Exam Problem (Concordia Midterm)

A cylindrical titanium alloy rod ($d_0 = 10.0\ \text{mm}$, $l_0 = 50.0\ \text{mm}$, $E = 107\ \text{GPa}$, $\nu = 0.34$) is pulled with an elastic tensile force of $F = 40.0\ \text{kN}$. Calculate:

(a) Engineering stress σ (in MPa), (b) Longitudinal extension Δl (in mm), (c) Change in diameter Δd (in mm), and (d) Shear modulus G (in GPa).

Zero-Skip Step-by-Step Solution

Step 1: Cross-Sectional Area and Stress

$$A_0 = \frac{\pi d_0^2}{4} = \frac{\pi (10.0\ \text{mm})^2}{4} = 25\pi\ \text{mm}^2 = 78.5398\ \text{mm}^2$$

$$\sigma = \frac{F}{A_0} = \frac{40000\ \text{N}}{78.5398\ \text{mm}^2} = \mathbf{509.30}\ \text{MPa}$$

Step 2: Elongation Δl

$$\epsilon_z = \frac{\sigma}{E} = \frac{509.30\ \text{MPa}}{107000\ \text{MPa}} = 0.0047598$$

$$\Delta l = \epsilon_z \times l_0 = 0.0047598 \times 50.0\ \text{mm} = \mathbf{0.238}\ \text{mm}$$

Step 3: Contraction in Diameter Δd

$$\epsilon_x = -\nu \epsilon_z = -(0.34) \times (0.0047598) = -0.0016183$$

$$\Delta d = \epsilon_x \times d_0 = (-0.0016183) \times 10.0\ \text{mm} = \mathbf{-0.0162}\ \text{mm}$$

Step 4: Shear Modulus G

$$G = \frac{E}{2(1+\nu)} = \frac{107\ \text{GPa}}{2(1+0.34)} = \frac{107}{2.68} = \mathbf{39.93}\ \text{GPa}$$

5.2 Ductility (%EL, %RA) & Modulus of Resilience (U_r)

$$\boxed{\%\text{EL} = \frac{l_f - l_0}{l_0} \times 100\% \qquad \%\text{RA} = \frac{A_0 - A_f}{A_0} \times 100\% \qquad U_r = \frac{\sigma_y^2}{2E}}$$

Past Exam Problem (Concordia Midterm)

A structural steel cylindrical specimen ($d_0 = 12.8\ \text{mm}$, $l_0 = 50.8\ \text{mm}$) has $\sigma_y = 450\ \text{MPa}$ and $E = 207\ \text{GPa}$. At fracture, $l_f = 67.2\ \text{mm}$ and neck diameter $d_f = 8.5\ \text{mm}$. Calculate (a) U_r in kJ/m^3, (b) $\%\text{EL}$, and (c) $\%\text{RA}$.

Zero-Skip Step-by-Step Solution

Step 1: Modulus of Resilience U_r

$$U_r = \frac{(450 \times 10^6\ \text{Pa})^2}{2 \times (207 \times 10^9\ \text{Pa})} = \frac{2.025 \times 10^{17}}{4.14 \times 10^{11}} = 489130\ \text{J/m}^3 = \mathbf{489.1}\ \text{kJ/m}^\mathbf{3} = \mathbf{489.1}\ \text{kPa}$$

Step 2: Percent Elongation $\%\text{EL}$

$$\%\text{EL} = \frac{67.2 - 50.8}{50.8} \times 100\% = \frac{16.4}{50.8} \times 100\% = \mathbf{32.28\%}$$

Step 3: Percent Reduction in Area $\%\text{RA}$

$$A_0 = \frac{\pi}{4}(12.8)^2 = 128.68 \text{ mm}^2 \qquad A_f = \frac{\pi}{4}(8.5)^2 = 56.75 \text{ mm}^2$$

$$\%\text{RA} = \frac{128.68 - 56.75}{128.68} \times 100\% = \frac{71.93}{128.68} \times 100\% = \mathbf{55.90\%}$$

5.3 True Stress, True Strain & Hollomon's Law

$$\boxed{\sigma_T = \sigma(1+\epsilon) \qquad \epsilon_T = \ln(1+\epsilon) \qquad \sigma_T = K\epsilon_T^n}$$

Past Exam Problem (Concordia Midterm)

Tensile plastic flow data for an annealed brass alloy prior to necking shows:

- Point 1: $\sigma_{T1} = 300$ MPa at $\epsilon_{T1} = 0.10$

- Point 2: $\sigma_{T2} = 410$ MPa at $\epsilon_{T2} = 0.25$

1. Determine the strain-hardening exponent n and strength coefficient K (in MPa).

2. Predict the true stress σ_T required to produce a true strain of $\epsilon_T = 0.18$.

3. If an engineering strain of $\epsilon = 0.12$ is applied, determine the corresponding true stress σ_T.

Zero-Skip Step-by-Step Solution

Step 1: Calculate n **and** K

$$n = \frac{\ln(\sigma_{T2}/\sigma_{T1})}{\ln(\epsilon_{T2}/\epsilon_{T1})} = \frac{\ln(410/300)}{\ln(0.25/0.10)} = \frac{\ln(1.36667)}{\ln(2.50)} = \frac{0.31238}{0.91629} = \mathbf{0.3409} \approx \mathbf{0.341}$$

$$K = \frac{\sigma_{T1}}{\epsilon_{T1}^n} = \frac{300 \text{ MPa}}{(0.10)^{0.3409}} = \frac{300}{0.45614} = \mathbf{657.7} \text{ MPa}$$

Step 2: Predict True Stress at $\epsilon_T = 0.18$

$$\sigma_T = 657.7 \times (0.18)^{0.3409} = 657.7 \times 0.55835 = \mathbf{367.23} \text{ MPa}$$

Step 3: Convert Engineering Strain to True Strain & True Stress

$$\epsilon_T = \ln(1 + 0.12) = \ln(1.12) = 0.11333$$

$$\sigma_T = 657.7 \times (0.11333)^{0.3409} = 657.7 \times 0.47648 = \mathbf{313.38} \text{ MPa}$$

> **Why This Method Works**: Hollomon's law models dislocation multiplication and entanglement during cold work. The slope of $\ln \sigma_T$ vs $\ln \epsilon_T$ is constant (n) throughout uniform plastic deformation.

>

> **Exam Trap:** Using Hollomon's equation beyond the ultimate tensile strength (necking). Once necking initiates,

deformation is non-uniform and hydrostatic triaxial stress develops.

5.4 Factor of Safety & Safe Working Stress

$$\boxed{\sigma_w = \frac{\sigma_y}{N} \qquad A_{\min} = \frac{F}{\sigma_w} = \frac{F \cdot N}{\sigma_y}}$$

Past Exam Problem (Concordia Midterm)

A structural tie-rod forged from heat-treated alloy steel (yield strength $\sigma_y = 750\ \text{MPa}$) must safely support a static cyclic tensile load of $F = 65.0\ \text{kN}$ ($65000\ \text{N}$) with a design factor of safety of $N = 2.50$. Calculate (a) the maximum safe working stress σ_w, and (b) the minimum circular rod diameter $d_{\min}$ in millimeters.

Zero-Skip Step-by-Step Solution

Step 1: Calculate Safe Working Stress σ_w

$$\sigma_w = \frac{\sigma_y}{N} = \frac{750\ \text{MPa}}{2.50} = \mathbf{300.0}\ \text{MPa} = 300.0\ \text{N/mm}^2$$

Step 2: Calculate Minimum Cross-Sectional Area $A_{\min}$

$$A_{\min} = \frac{F}{\sigma_w} = \frac{65000\ \text{N}}{300.0\ \text{N/mm}^2} = 216.667\ \text{mm}^2$$

Step 3: Calculate Minimum Rod Diameter $d_{\min}$

$$A = \frac{\pi d^2}{4} \implies d_{\min} = \sqrt{\frac{4A_{\min}}{\pi}} = \sqrt{\frac{4 \times 216.667\ \text{mm}^2}{\pi}} = \sqrt{\frac{866.667}{3.14159}} = \sqrt{275.869} = \mathbf{16.61}\ \text{mm}$$

(A standard nominal rod diameter of $\geq 17.0\ \text{mm}$ must be specified).